

Samsung 9100 Pro 8TB Gen5 SSD

Not afraid to take up space!

₹1,59,990



DESIGN AND BUILD

At a glance, the Samsung 9100 Pro does not try to look flashy. The overall design is clean and functional, which is something I personally appreciate. Lower-capacity models in the lineup are single-sided, with all the components mounted on one side of the PCB. But the 8 TB model is different.

To achieve this capacity, Samsung has gone with a double-sided PCB for the first time in its consumer SSD lineup. That is because the highest-capacity NAND packages currently top out at 2 TB, so fitting 8 TB into an M.2 2280 form factor requires NAND chips on both sides.

The heatsink version I tested is clearly designed with practicality in mind. The updated heatsink is slim, compact, and finished with subtle grooves that give it a professional, almost industrial look. It does not scream for attention, but it feels purpose-built.

One thing worth noting is thickness. The 8 TB

model is slightly thicker than the 1 TB to 4 TB variants. While the smaller models measure around 6.5 mm on top and 1.5 mm on the bottom, the 8 TB version measures roughly 8 mm on top and 2.45 mm underneath. Including the PCB, the total thickness comes to about 11.25 mm, compared to roughly 8.88 mm on the lower-capacity drives. Despite this, I faced absolutely no issues installing it on my PC.

SOFTWARE EXPERIENCE

The SSD comes equipped with the Samsung Magician software, which gives you everything you could ever realistically need. This includes drive information, health monitoring, firmware updates, built-in benchmarking, diagnostics, and secure erase tools.

Samsung's Data Migration software is also extremely useful. It allows you to clone data from one Samsung SSD to another, even when the source drive is

As someone who reviews tech and is also an avid gamer with hundreds of titles sitting in my Steam library, storage is one of those components I can never seem to have enough of. Between modern AAA games that routinely exceed the 100 GB mark, large project files, and the need to keep multiple installations ready to go, running out of space is no longer an inconvenience. It is a workflow breaker. And that's exactly why, when it comes to SSDs, I have always leaned towards the 'larger the better' school of thought.

When I got a chance to review the 8TB heatsink variant of the Samsung 9100 Pro SSD, I was quite excited to see how it turned out. First off, this is not an SSD designed for casual upgrades or modest game libraries. It is built for users who want everything in one place, fast and always accessible. With PCIe Gen5 speeds and an unusually high capacity packed into a single M.2 drive, the 9100 Pro aims to remove storage limitations altogether. But it is also quite expensive when compared to other SSDs. Now, the question is, should you invest in it? Read on, and let's try to find the answer.

TEST SYSTEM SETUP

I tested the Samsung 9100 Pro 8 TB on

my PC, which has the following configuration:

- Motherboard: Asus ROG Strix X870E-E
- Processor: AMD Ryzen 7 9800X3D
- Memory: 32 GB G.Skill Trident Z5 DDR5-6000
- Graphics card: Nvidia RTX 4070 Ti Super
- Operating system: Windows 11

For context, I have previously had the Samsung 990 Pro 2 TB Gen4 SSD installed on my system as the primary storage drive. Hence, I am no stranger to Samsung SSDs and have a fair idea of what I can expect from this massive storage upgrade.

Now, the Samsung 9100 Pro series starts at 1 TB and goes all the way up to 8 TB. I am testing the top-end variant here, which is very clearly aimed at professionals rather than everyday users. This is a PCIe Gen5 SSD, which means it can theoretically deliver far higher bandwidth than Gen4 drives. It is also fully backwards compatible, so you can run it in a Gen4 slot, but you will be capped at Gen4 speeds.

To actually experience what this SSD is capable of, a modern Gen5 platform is essential. On older systems, you are paying a premium without unlocking its full potential.

